ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

COMPUTER HARDWARE AND SOFTWARE

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

NC(V) Information Technology and Computer Science, Level 3. Section A: objective questions (30) · Section B: system unit components, storage and memory, number systems, operating systems and installation, maintenance and troubleshooting (70).

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL calculations and conversions.
5. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 The firmware that tests the hardware when a computer is switched on and starts the boot process is the …

A CMOS battery. B device driver. C kernel. D BIOS / UEFI.

1.1.2 The test the computer performs on its hardware at start-up is called the …

A FAT. B POST. C RAID. D GUI.

1.1.3 Which expansion slot is used for modern graphics cards?

A AGP B ISA C SATA D PCI Express x16

1.1.4 Very fast memory built into the CPU that stores frequently used instructions is …

A virtual memory. B cache memory. C ROM. D flash memory.

1.1.5 The binary number 1011 is equal to the decimal number …

A 9. B 10. C 11. D 13.

1.1.6 Which interface is used to connect an internal SSD or hard drive to a modern motherboard?

A SATA B VGA C PS/2 D RJ45

1.1.7 The file system normally used for Windows system drives is …

A FAT16. B NTFS. C ext4. D HFS+.

1.1.8 Thermal paste is applied between the …

A RAM and its slot. B power supply and case.
C hard drive and cable. D CPU and its heat sink.

1.1.9 Software that lets the operating system communicate with a hardware device is a …

A device driver. B compiler.
C utility. D macro.

1.1.10 A computer beeps repeatedly at start-up and nothing appears on the screen. The FIRST thing to check is …

A the printer cable.
B the antivirus.
C the RAM modules are seated properly.
D the screen saver settings.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 The power supply unit converts AC from the wall socket into DC for the components.

1.2.2 DDR4 memory modules can be inserted into DDR3 slots.

1.2.3 Defragmenting an SSD regularly improves its lifespan.

1.2.4 The hexadecimal number F is equal to decimal 15.

1.2.5 A 64-bit operating system can use more than 4 GB of RAM.

1.2.6 Virtual memory uses part of the storage drive as if it were RAM.

1.2.7 An uninterruptible power supply (UPS) keeps the computer running for a short time during a power failure.

1.2.8 Partitioning divides a physical drive into separate logical drives.

1.2.9 Proprietary software can be freely modified by anyone.

1.2.10 A higher clock speed (GHz) generally means the CPU can execute more instructions per second.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	Chip that controls communication between the CPU, memory and other components	A	HDMI
1.3.2	Small battery-powered memory that stores BIOS settings and the date	B	chipset
1.3.3	Several drives combined for speed or for protection against failure	C	antivirus
1.3.4	The core of the operating system that manages memory and processes	D	firmware
1.3.5	Program used to detect and remove malicious software	E	RAID
1.3.6	Restores the operating system to an earlier working state	F	cores
1.3.7	Software installed into hardware such as a router or BIOS chip	G	system restore
1.3.8	Number of processing units inside one CPU	H	CMOS
1.3.9	Interface that connects a monitor and carries digital video and sound	I	product key (licence)
1.3.10	Key that proves software was legally bought	J	kernel

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2: SYSTEM UNIT COMPONENTS

2.1 Name FIVE components found on a motherboard. (5)

2.2 Describe the machine cycle (fetch–decode–execute) of the CPU. (6)

2.3 Compare a hard disk drive (HDD) and a solid-state drive (SSD) under the headings speed, durability and cost per gigabyte. (6)

2.4 Explain why adding more RAM can make a computer faster. (3)

[20]

QUESTION 3: NUMBER SYSTEMS AND DATA REPRESENTATION

3.1 Convert decimal 173 to binary. Show your working. (3)

3.2 Convert binary 11010110 to hexadecimal. (2)

3.3 Convert hexadecimal 2F to decimal. (2)

3.4 Add the binary numbers 1011 0110 and 0011 1011. Give the answer in binary. (3)

3.5 What is ASCII, and why is Unicode used today instead of ASCII for many applications? (3)

3.6 A 500 GB drive is advertised, but Windows shows about 465 GB. Explain why. (2)

[15]

QUESTION 4: OPERATING SYSTEMS AND SOFTWARE INSTALLATION

> A college computer lab has 30 computers that must be set up with a new operating system, office software and programming tools before the term starts.

4.1 List FIVE steps, in order, to install an operating system on a new computer. (5)

4.2 Give TWO reasons why the technician should check the minimum system requirements before installing software. (2)

4.3 Name TWO ways the technician can install the same software on 30 computers quickly. (2)

4.4 Distinguish between open-source and proprietary software. Give ONE example of each. (4)

4.5 Explain the purpose of user accounts and permissions in the lab. (2)

[15]

QUESTION 5: MAINTENANCE AND TROUBLESHOOTING

5.1 Name FOUR preventive maintenance tasks for computers and say how often each should be done. (8)

5.2 List the SIX steps of the troubleshooting process in the correct order. (6)

5.3 For EACH symptom, give a likely cause and a solution: (a) the computer switches off by itself after a few minutes of heavy use; (b) the computer is very slow and pop-up adverts keep appearing. (4)

5.4 Explain how e-waste (old computers) should be disposed of, and why. (2)

[20]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER